SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: STATISTICS WITH R PROGRAMMING **COURSE CODE: ITA0404**

COURSE OUTCOME COVERED

CO2 -- ASSESSMENT TOOL 2

CO2: Implement looping constructs, control structures, and recursive functions in R to solve computational problems with efficient program logic.Analyse the Correlation between the parameters in a dataset and establish the analysis (k3)

Assessment Tool Weightage: **40%**

QUESTIONS: 5 **Total Marks: 50** **Duration: 1 Hour**

Coding Challenge

Code of Conduct:

I J.Raghavi Reddy (192525158) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: j.RaghaviReddy

Reg. No: 192525158

RUBRICS FOR CALCULATION

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30						
Accuracy of Answer	25						

Application of Knowledge	20						
Completeness of Response	15						
Clarity & Presentation	10						
Total per Question	10 Marks	/10	/10	/10	/10	/10	**/ 50**

SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

1. Employee Bonus Calculator

Write an R function bonus() that accepts an employee's **salary** and **performance rating**. If rating ≥ 4, give a 10% bonus.
Otherwise, give a 5% bonus.
Use if-else.
Return the bonus amount.
Set the default rating as 3.

2. Shopping Discount Calculator

Write an R function discount() that accepts the **purchase amount** and **membership status**.
If the amount is greater than ₹5,000 **and** the customer is a member, give a 10% discount.
Otherwise, give no discount.
Use Boolean operators.
Return the discount amount.
Set the default membership status as FALSE.

3. Loan Eligibility Checker

Write an R function loan_check() that accepts **age** and **income**.
A customer is eligible for a loan if:
Age is 21 or above **and**
Income is ₹30,000 or more. Use if-else and Boolean operators.
Return "Eligible" or "Not Eligible".

1. Electricity Bill Calculator

Write an R function electricity_bill() that accepts the number of **units consumed**.
Up to 100 units → ₹2 per unit.
Above 100 units → ₹3 per unit.
Add a default fixed charge of ₹100.

Return the total bill.
If units are negative, return "Invalid Units".

2. ATM Withdrawal Checker

Write an R function withdraw() that accepts **balance** and **amount**.

Check whether the amount is greater than 0.
Check whether the amount is less than or equal to the balance.
If both conditions are true, return the remaining balance.
Otherwise, return "Transaction Failed".
Set the default withdrawal amount as 1000.

Topics covered: if-else, arithmetic operators, Boolean operators, default arguments, and return values.

Bottom of Form

CO2 AT-2

NAME: J.RaghaviReddy FACULTY: Dr. G.Kumaresan

REG.NO: 192525158 COURSE CODE:ITA040 4

COURSE NAME: Statistics With R Programming

1.Employee Bonus Calculator

Aim: To write an R function bonus() to calculate an employee's bonus based on salary and performance rating using if-else**.**

Code:

```
bonus <- function(salary, rating =
3) {   if (rating >= 4)
{     bonus_amount <- salary * 0.10

  } else {     bonus_amount <-
salary * 0.05

  }


  return(bonus_amount)

}


salary <- 50000
rating <- 4


result <- bonus(salary, rating)
print(result)
```

Input:

Salary = 50000
Rating = 4

Output:

[1] 5000

Result: The R program was successfully executed, and the employee bonus was calculated based on the performance rating. For a salary of ₹50,000 and rating 4, the bonus obtained is ₹5,000.

2. Shopping Discount Calculator

Aim: To write an R function discount() to calculate the discount amount based on purchase amount and membership status using Boolean operators.

Code:

```
discount <- function(amount, member = FALSE)
{   if (amount > 5000 && member == TRUE)
{     discount_amount <- amount * 0.10

  } else
{     discount_amount <-
0

  }


  return(discount_amount)

}


amount <- 8000
member <- TRUE


result <- discount(amount, member)
print(result)
```

Input:

Purchase Amount =
8000 Member = TRUE

Output:

[1] 800

Result:

The R program was successfully executed, and the discount was calculated based on the purchase amount and membership status. For a purchase amount of ₹8,000 and membership status TRUE, the discount obtained is ₹800.

3. Loan Eligibility Checker

Aim: To write an R function loan_check() to determine whether a customer is eligible for a loan based on age and income.

Code:

```
loan_check <- function(age, income)
{   if (age >= 21 && income >=
30000) {     return("Eligible")

  } else
{     return("Not
Eligible")

  }

}


age <- 25
income <-
40000


result <- loan_check(age, income)
print(result)
```

Input:

Age = 25

Income = 40000

Output:

[1] "Eligible"

Result:

The R program was successfully executed, and the customer's loan eligibility was checked based on age and income. For age 25 and income ₹40,000, the result is "Eligible".

4. Electricity Bill Calculator

Aim: To write an R function electricity_bill() to calculate the total electricity bill based on the number of units consumed.

Code:

```
electricity_bill <- function(units)
{   fixed_charge <- 100


  if (units < 0)
{     return("Invalid
Units")   } else if (units
<= 100) {
    bill <- (units * 2) + fixed_charge
  } else {     bill <- (units * 3) +
fixed_charge
  }


  return(bill)
}


units <- 150


result <- electricity_bill(units)
print(result)
```

Input:

Units Consumed = 150

Output:

[1] 550

Result:

The R program was successfully executed, and the electricity bill was calculated based on the units consumed. For 150 units, including the fixed charge of ₹100, the total bill obtained is ₹550.

5. ATM Withdrawal Checker

Aim: To write an R function withdraw() to check whether an ATM withdrawal is possible based on the account balance and withdrawal amount. **Code:**

```
withdraw <- function(balance, amount =
1000) {   if (amount > 0 && amount <=
balance) {     remaining_balance <- balance -
amount     return(remaining_balance)
  } else
{     return("Transaction
Failed")
  }
}


balance <- 5000
amount <- 1000


result <- withdraw(balance, amount)
print(result)
```

Input:

Balance = 5000

Withdrawal Amount = 1000

Output:

[1] 4000

Result:

The R program was successfully executed, and the ATM withdrawal was checked based on the available balance and withdrawal amount. For a balance of ₹5,000 and withdrawal of ₹1,000, the remaining balance is ₹4,000.